

Calculus I Lesson 29

1) Find $\lim_{x \rightarrow \infty} \frac{5x+3}{x-1}$

a) $\frac{5x+3}{x-1} = \frac{5x/x + 3/x}{x/x - 1/x} = \underline{\quad ? \quad}$

b) $\lim_{x \rightarrow \infty} \frac{5x+3}{x-1} = \underline{\quad ? \quad}$

2) Find $\lim_{x \rightarrow \infty} \frac{1}{x^2 - 4}$

a) $\frac{1}{x^2 - 4} = \frac{1/x^2}{x^2/x^2 - 4/x^2} = \underline{\hspace{2cm}}?$

b) $\lim_{x \rightarrow \infty} \frac{1}{x^2 - 4} = \underline{\hspace{2cm}}?$

3) Find $\lim_{x \rightarrow \infty} \frac{4x^2 + 3}{x^3 - 1}$

a) $\frac{4x^2 + 3}{x^3 - 1} = \frac{4x^2/x^3 + 3/x^3}{x^3/x^3 - 1/x^3} = \underline{\hspace{2cm}}?$

b) $\lim_{x \rightarrow \infty} \frac{4x^2 + 3}{x^3 - 1} = \underline{\hspace{2cm}}?$

4) Find $\lim_{x \rightarrow \infty} \left(\frac{5x^{1/2} + 3x}{x^2 - 3} \right)$

a) $\frac{5x^{1/2} + 3x}{x^2 / x^2 - 3} = \frac{5x^{1/2} / x^2 + 3x / x^2}{x^2 / x^2 - 3 / x^2} = \underline{\quad ? \quad}$

Hint: $5x^{1/2} / x^2 = 5x^{1/2-2} = 5x^{-3/2} = \frac{5}{x^{3/2}};$

$$\lim_{x \rightarrow \infty} x^{0.1} = \infty; \quad \lim_{x \rightarrow \infty} x^{1/2} = \infty; \quad \lim_{x \rightarrow \infty} x^{0.001} = \infty;$$

$$\lim_{x \rightarrow \infty} x^{-0.1} = \lim_{x \rightarrow \infty} \frac{1}{x^{0.1}} = 0; \quad \lim_{x \rightarrow \infty} x^{-1/2} = \lim_{x \rightarrow \infty} \frac{1}{x^{1/2}} = 0; \quad \lim_{x \rightarrow \infty} x^{-0.001} = \lim_{x \rightarrow \infty} \frac{1}{x^{0.001}} = 0;$$

Hence, $\lim_{x \rightarrow \infty} x^a (>0) = \infty$ and $\lim_{x \rightarrow \infty} x^b (<0) = 0$

b) $\lim_{x \rightarrow \infty} \left(\frac{5x^{1/2} + 3x}{x^2 - 3} \right) = \underline{\quad ? \quad}$

5) Find $\lim_{x \rightarrow \infty} \left(\frac{x^{2/3} + 6x}{x^3 - 7} \right)$

a) $\frac{x^{2/3} + 6x}{x^3 - 7} = \frac{x^{2/3}/x^3 + 6x/x^3}{x^3/x^3 - 7/x^3} = \underline{\hspace{2cm}}?$

Hint: $x^{2/3}/x^3 = x^{2/3-3} = x^{-7/3} = \frac{1}{x^{7/3}};$

b) $\lim_{x \rightarrow \infty} \left(\frac{x^{2/3} + 6x}{x^3 - 7} \right) = \underline{\hspace{2cm}}?$

Hint: $\lim_{x \rightarrow \infty} x^a (>0) = \infty$ and $\lim_{x \rightarrow \infty} x^b (<0) = 0$

6) Find $\lim_{x \rightarrow \infty} \left(\frac{x^{-1/3} + 3x}{x^2 - 4x} \right)$

a) $\frac{x^{-1/3} + 3x}{x^2 - 4x} = \frac{x^{-1/3} / x^2 + 3x / x^2}{x^2 / x^2 - 4x / x^2} = \underline{\hspace{2cm}} ?$

Hint: $x^{-1/3} / x^2 = x^{-1/3-2} = x^{-7/3} = \frac{1}{x^{7/3}};$

b) $\lim_{x \rightarrow \infty} \left(\frac{x^{-1/3} + 3x}{x^2 - 4x} \right) = \underline{\hspace{2cm}} ?$

Hint: $\lim_{x \rightarrow \infty} x^a (>0) = \infty$ and $\lim_{x \rightarrow \infty} x^b (<0) = 0$

